

This project, titled **StudentPerformancePrediction-ML**, is a machine learning initiative focused on predicting student grades based on various factors using a provided CSV dataset.

The core goal is to determine which factors influence student marks and to use different machine learning classifiers to predict student performance (High, Medium, or Low grade class).

Project Components and Methodology

1. Data Source

The project uses a dataset from the **AI-Data.csv** file. This dataset contains information about students from various nationalities and grade levels, along with several determining factors of student engagement and parent involvement.

The columns in the dataset include:

- **Demographic/Contextual:** gender, Nationality, PlaceofBirth, StageID, GradeID, SectionID, Topic, Semester, Relation (Parent relation to student, i.e., Father/Mum).
- **Engagement Metrics (Numerical):** raisedhands (number of times student raised hand), VisitedResources (number of times student visited educational resources), AnnouncementsView (number of times student viewed announcements), Discussion (number of times student participated in discussion groups).
- **Parent/Absence:** ParentAnsweringSurvey, ParentschoolSatisfaction, StudentAbsenceDays (Under-7 or Above-7 days).
- **Target Variable:** Class (the student's performance or grade class, categorized as **L** for Low, **M** for Medium, or **H** for High).

2. Analysis and Visualization (Exploratory Data Analysis - EDA)

The Python script includes an interactive section for performing exploratory analysis on the data by generating various graphs.

The visualizations implemented are:

- **Correlation Heatmap:** To visualize the linear correlation between the numerical variables in the dataset.
- **Marks Class Count Graph:** Total count of students in each class (L, M, H).
- **Comparative Graphs:** Count plots of the Class (L, M, H) against several categorical variables:
 - Semester-wise Graph
 - Gender-wise Graph
 - Nationality-wise Graph
 - Grade-wise Graph
 - Section-wise Graph
 - Topic-wise Graph

- Stage-wise Graph
- Absent Days-wise Graph

3. Machine Learning Models (Prediction)

The project trains and evaluates several machine learning classification models to predict the student's final `Class`.

Pre-processing and Feature Selection:

A significant number of columns are dropped before training the models, likely to focus on the most impactful features or to simplify the initial model. The dropped columns are: gender, StageID, GradeID, Nationality, PlaceofBirth, SectionID, Topic, Semester, Relation, ParentschoolSatisfaction, and ParentAnsweringSurvey.

The remaining features are processed:

1. Categorical features are encoded using **Label Encoding** (e.g., `StudentAbsenceDays` and the target variable `Class` are converted to numerical values).
2. Numerical columns are scaled using **StandardScaler** (though the specific list of columns used for training is small based on the feature dropping).

Models Used:

The Python code trains and reports the accuracy for five different classification models:

1. **Decision Tree Classifier** (`modelD`)
2. **Random Forest Classifier** (`modelR`)
3. **Perceptron Classifier** (`modelP`) - likely referring to a type of linear model or a single layer of the neural network.
4. **Linear Discriminant Analysis (LDA) / Linear Regression** (`modelL`) - The variable `modelL` is used for `lm.LogisticRegression()`, which is a Logistic Regression model (a classifier).
5. **Multi-layer Perceptron (MLP) Classifier** (`modelN`) - A basic Neural Network model.

The project also provides a final interactive section to test the trained models with a specific user-defined input.